

Semantic navigation from live robot perception

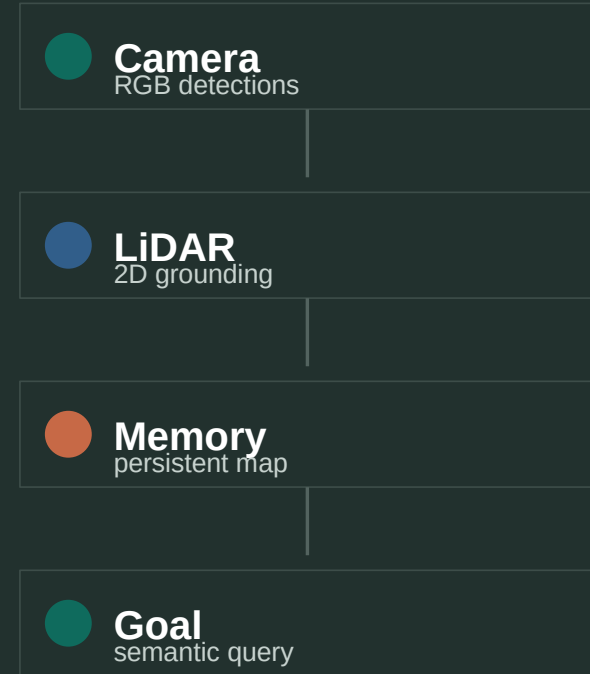
A layered ROS 2 stack for building queryable semantic maps from camera, LiDAR, TF, and memory.

no mock detections

semantic memory

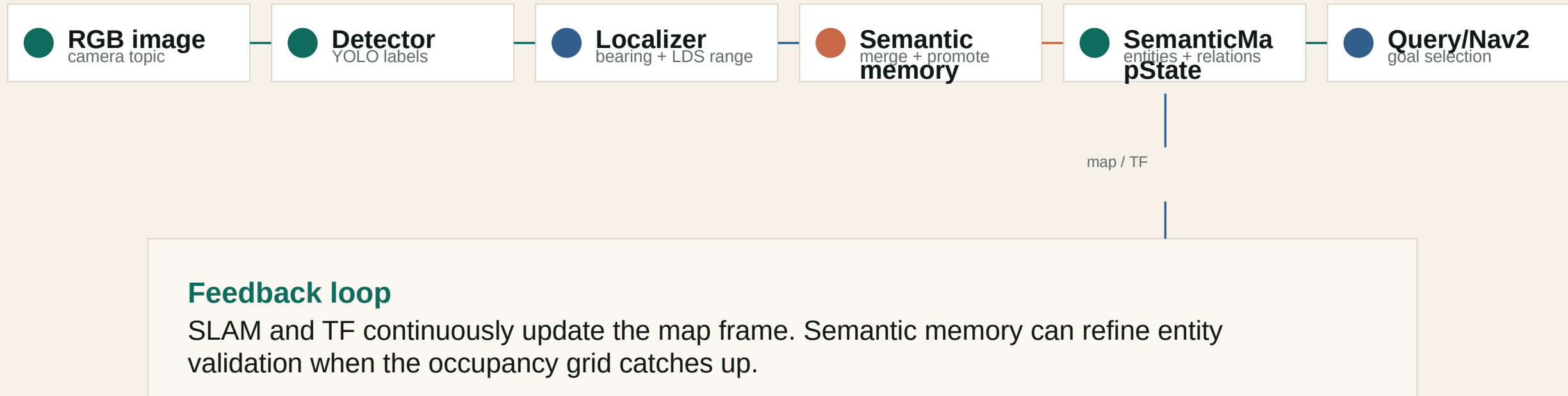
Nav2 integration

The system makes semantic navigation a runtime product of robot observation: what the robot sees, grounds, remembers, and relates becomes the map that downstream navigation uses.



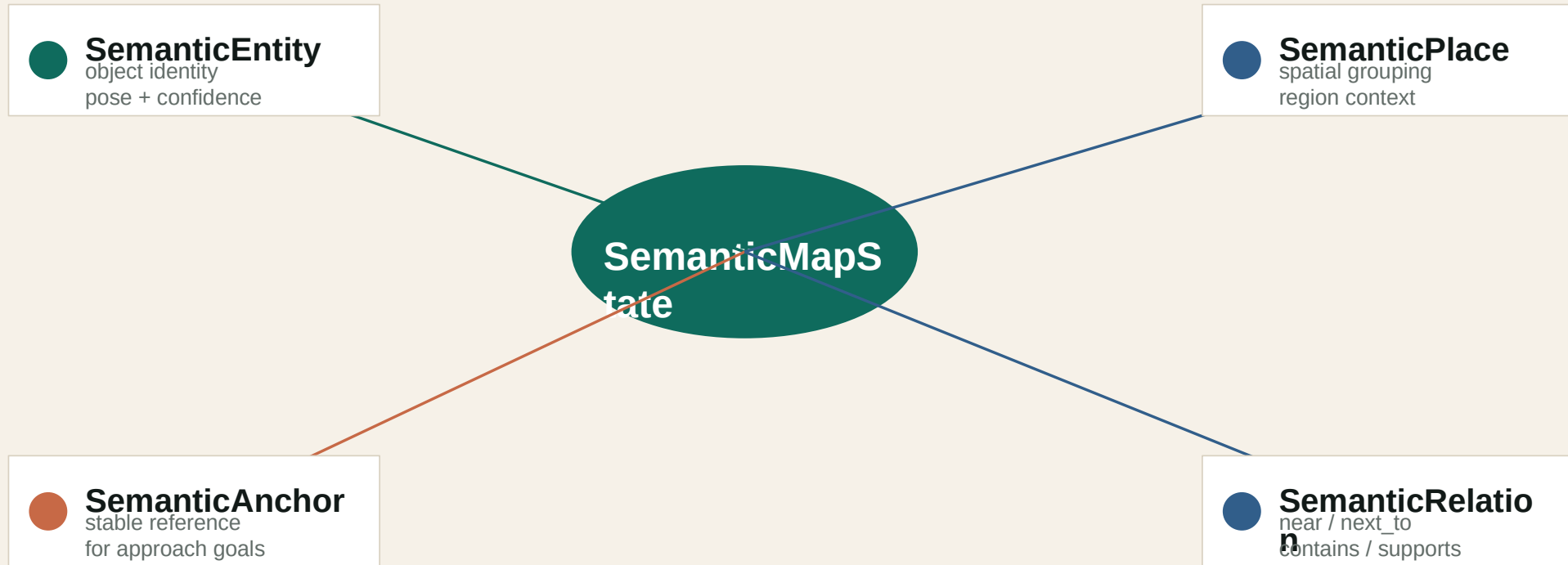
End-to-end flowchart

The system converts pixels into semantic goals through explicit ROS 2 stages.



Semantic map data model

The semantic nav dependency is structured around a hierarchy of map concepts.

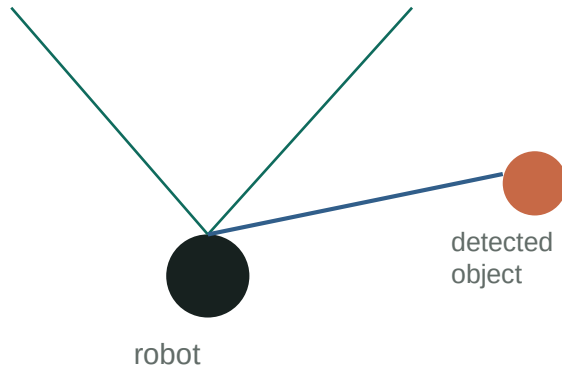


Entities are not isolated points; they are connected into places, anchors, and relations that queries can reason over.

Grounding real detections

Camera detections become usable only after robot-native spatial grounding.

Perception geometry



occupancy_grid_validated

object point is inside the map and attached to obstacle evidence

occupancy_grid_pending

real LiDAR-grounded object, but local obstacle evidence is not ready yet

map_bounds_pending

object is real and grounded, but SLAM bounds have not expanded to it yet

Semantic memory state machine

Repeated real observations are merged before becoming navigation targets.



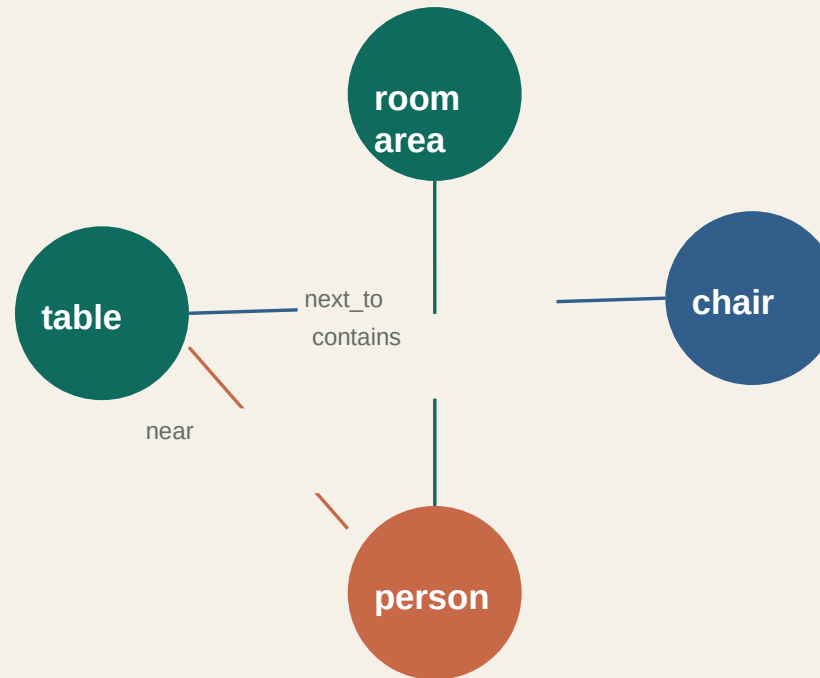
The memory node keeps provenance with each promoted landmark, so downstream navigation can distinguish validated entities from provisional ones.

Relationship mapping

The relationship layer turns objects into a navigable scene graph.

Label mapping

Detector labels stay separate from semantic names. A detector label like bench can resolve to the user-facing concept table.

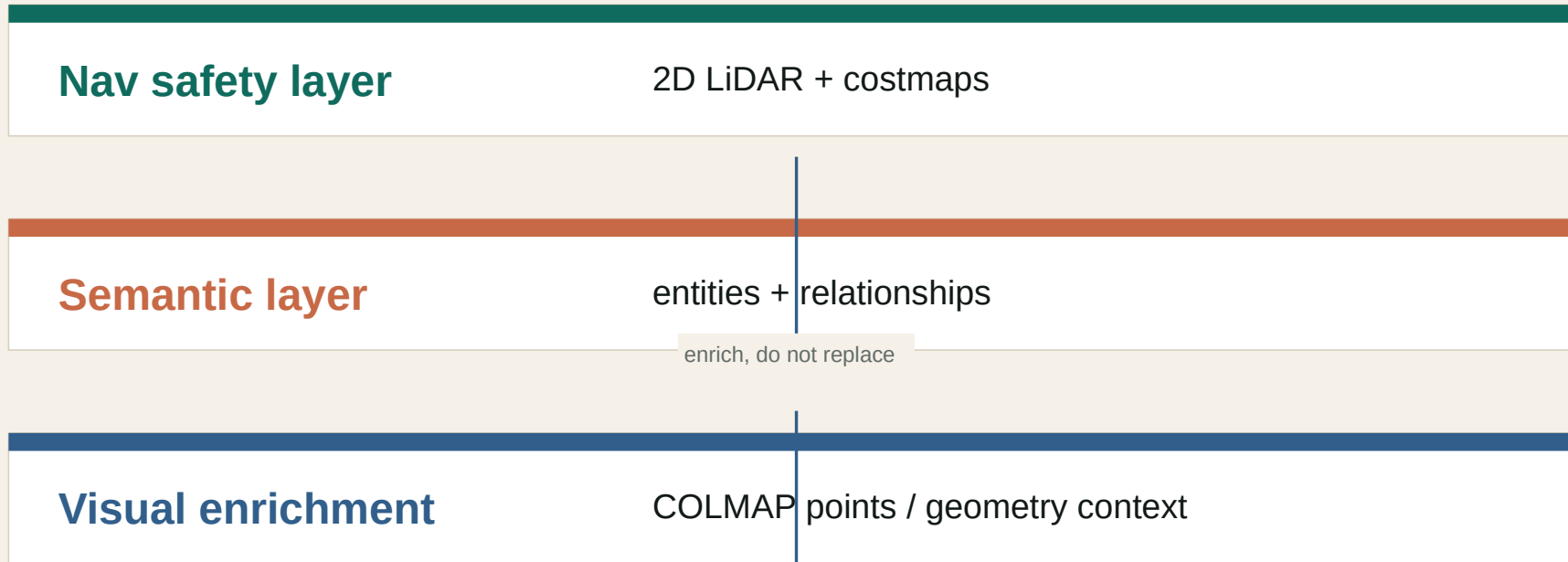


Vocabulary choice

Use robust relations like near and next_to instead of camera-relative left/right where viewpoint is unstable.

COLMAP enrichment for limited sensors

The visual reconstruction layer adds context without replacing robot safety grounding.



For 2D LiDAR robots, COLMAP helps recover visual geometry that the scan plane cannot see, while motion planning remains tied to the robot's validated obstacle model.

Real robot integration

The system was prepared for a TurtleBot3-style ROS 2 deployment path.



LDS-02 bringup

real /scan source



USB camera path

real image stream



Nav2 lifecycle

active planner/controller/BT



Motion safety

unarmed until explicit test



Semantic route

memory publishes /semantic_map/state

Important real-world fix

When the detected table/bench was just outside the current SLAM map bounds, the old memory path dropped it. The new path keeps it as a real LiDAR-grounded provisional landmark until map validation catches up.

What changed here

The implementation is now organized around real full-stack runs, not demos or fake artifacts.

Removed mock artifacts

No fake COLMAP points, bootstrap people, or hardcoded semantic detections.

Routed semantic state

Persistent semantic memory now publishes the native SemanticMapState used by query.

Added provisional landmarks

Real LiDAR-grounded entities survive temporary occupancy-grid/map-bound gaps.

Validated packages

Coordinator and semantic-map packages build and test cleanly.

Result

A semantic navigation stack where meaning is built from live robot observations, not predefined scene knowledge.

The robot can move from “I see something” to “I understand what it is, where it is, how it relates to the scene, and whether it is safe to navigate

Built around: live perception, LiDAR grounding, persistent semantic memory, relationship mapping, COLMAP enrichment, and ROS 2/Nav2 integration.